EE4-66 Topics in Large Dimensional Data Processing

Instructions for Candidates

Answer all questions. Each question carries 20 marks.

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1. (Convex Optimisation) For simplicity, it is assumed that $\text{domain}(f) = \mathbb{R}^n$ for any given function $f : \mathbb{R}^n \to \mathbb{R}$.

   (a) What is the definition that a function $f : \mathbb{R}^n \to \mathbb{R}$ is convex? [2]

   (b) Let $p : \mathbb{R}^n \to \mathbb{R}$ be a norm. Prove that p is convex. [3]

   (c) Let $f(x) = \frac{1}{2}(y - x)^2 + \lambda |x|$ where $y \in \mathbb{R}$ and $\lambda \in \mathbb{R}^+$ are given constants.

      i). What is the second condition for convexity? (Proof is not needed.) Use it to show that $\frac{1}{2}(y - x)^2$ is a convex function of x. [3]

      ii). Prove that a summation of convex functions is convex. Use this result to show that $f(x)$ is convex. [3]

      iii). Compute the subdifferential of $f(x)$, denoted by $\partial f(x)$, for all $x \in \mathbb{R}$. [3]

      iv). Find the $x^* \in \mathbb{R}$ that minimises $f(x)$. [3]

   (d) Let $f : \mathbb{R}^n \to \mathbb{R}$ be a convex function. Prove that if $\boldsymbol{x}^* \in \mathbb{R}^n$ is a local minimiser of f, then $\boldsymbol{x}^*$ is also a global minimiser of f. [3]

2. (Restricted Isometry Property)

(a) Given a matrix $\boldsymbol{A} \in \mathbb{R}^{m \times n}$, what is the definition that this matrix satisfies the Restricted Isometry Property (RIP) for $K \leq m$ and $\delta \in (0, 1)$? What is the definition of Restricted Isometry Constant (RIC) δ_K? [4]

(b) Show that for all $K < K' \leq m$, it holds that $\delta_K \leq \delta_{K'}$. [4]

(c) Suppose that $\boldsymbol{A}$ satisfies RIP with RIC $\delta_{2K} < 1$. Let $\mathcal{I}, \mathcal{J} \subset \{1, \cdots, n\}$ be two disjoint sets, i.e., $\mathcal{I} \bigcap \mathcal{J} = \phi$ where ϕ is the empty set. Assume that $|\mathcal{I}| \leq K$ and $|\mathcal{J}| \leq K$. Let $\boldsymbol{A}_{\mathcal{I}}$ and $\boldsymbol{A}_{\mathcal{J}}$ be the sub-matrices of $\boldsymbol{A}$ composed of the columns indexed by $\mathcal{I}$ and $\mathcal{J}$ respectively.

i). Use the fact that $|\langle \boldsymbol{A}_{\mathcal{I}} \boldsymbol{a}, \boldsymbol{A}_{\mathcal{J}} \boldsymbol{b} \rangle| \leq \delta_{2K} \|\boldsymbol{a}\|_2 \|\boldsymbol{b}\|_2$, $\forall \boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^K$, to prove that $\left\| \boldsymbol{A}_{\mathcal{I}}^T \boldsymbol{A}_{\mathcal{J}} \boldsymbol{b} \right\|_2 \leq \delta_{2K} \|\boldsymbol{b}\|_2$ where the superscript T denotes the matrix transpose. [4]

ii). Let $\boldsymbol{y} = \boldsymbol{A}_{\mathcal{I}} \boldsymbol{x}$ where $\boldsymbol{x} \in \mathbb{R}^{|\mathcal{I}|}$. Define

$$\hat{\mathcal{I}} = \left\{ K \text{ indices corresponding to the } K \text{ largest magnitudes of } \boldsymbol{A}^T \boldsymbol{y} \right\}.$$

This definitions suggests that $\left\| \boldsymbol{A}_{\hat{\mathcal{I}}}^T \boldsymbol{y} \right\|_2 \geq \left\| \boldsymbol{A}_{\mathcal{I}}^T \boldsymbol{y} \right\|_2$.
Prove that if $\delta_{2K} < \frac{1}{2}$ then $\hat{\mathcal{I}} \bigcap \mathcal{I} \neq \phi$. [4]

iii). Let $\boldsymbol{x}_{\hat{\mathcal{I}} \bigcap \mathcal{I}}$ be the sub-vector of $\boldsymbol{x}$ corresponding to $\boldsymbol{A}_{\hat{\mathcal{I}} \bigcap \mathcal{I}}$. Show that

$$\left\| \boldsymbol{A}_{\hat{\mathcal{I}}}^T \boldsymbol{y} \right\|_2 \leq \left\| \boldsymbol{A}_{\hat{\mathcal{I}} \bigcap \mathcal{I}}^T \boldsymbol{y} \right\|_2 + \delta_{2K} \|\boldsymbol{x}\|_2 , \tag{2.1}$$

and

$$\left\| \boldsymbol{A}_{\hat{\mathcal{I}} \bigcap \mathcal{I}}^T \boldsymbol{y} \right\|_2 \leq (1 + \delta_{2K}) \left\| \boldsymbol{x}_{\hat{\mathcal{I}} \bigcap \mathcal{I}} \right\|_2 + \delta_{2K} \|\boldsymbol{x}\|_2 . \tag{2.2}$$

Use the inequalities to show that

$$\left\| \boldsymbol{x}_{\hat{\mathcal{I}} \bigcap \mathcal{I}} \right\|_2 \geq \frac{1 - 3\delta_{2K}}{1 + \delta_{2K}} \|\boldsymbol{x}\|_2 . \tag{2.3}$$

[4]

3. (Maximum Clique Problem) We consider the graph $G(n, 1/2)$ with n nodes where there is an edge between two distinct nodes with probability $1/2$ independently of other pair of nodes. For such graphs, we would like to study the *maximum clique problem*. We call a *clique* in a graph a subset S of nodes such that any pair of nodes $i, j \in S$ is connected. Let w_n be the size of the largest clique in $G(n, 1/2)$, i.e. the clique with the most number of nodes.

   (a) We will first show that, for a given $\epsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}\left(w_n \geq (2+\epsilon)\log_2(n)\right) = 0. \tag{2.4}$$

   i). Show that

$$\mathbb{P}\left(w_n \geq k\right) \leq 2^{k\log_2(n) - k(k-1)/2}.$$

   Hint: Use the union bound and the fact that $\binom{n}{k} \leq n^k$. [3]

   ii). For $k = (2+\epsilon)\log_2(n)$ show that (2.4) holds. [3]

   (b) We now describe a greedy algorithm *greedy clique* that extracts a clique of size $(1-\epsilon)\log_2(n)$ which has half the size of the maximum clique.

   Greedy clique: Pick a vertex v_1 in $G(n, 1/2)$, then pick a random neighbour of v_1 that you add to the set S'. Continue adding nodes to S' that are picked at random from the nodes that are neighbours to all nodes in S', i.e. have an edge to any node that has been so far included in S', as long as this can be done.

   i). Explain that the nodes in S' form a a clique. [1]

   ii). Let $q_k = \mathbb{P}(\text{Greedy clique terminates with a clique of size } k)$. Show that

$$q_k \leq \binom{n}{k}(1 - \frac{1}{2^k})^{n-k}.$$

   Hint: Use the inequality $(1-x) \leq e^{-x}$. [5]

   iii). Let $k_0 = (1-\epsilon)\log_2(n)$, show that $q_{k_0} = \exp(-Cn^{\epsilon}))$, for a given constant C (independent of n and ϵ) and large n. [4]

   iv). Using the fact that the sequence q_k is increasing for $k \in \{1, \ldots, \lceil \frac{n-1}{2} \rceil\}$, show that

$$\lim_{n\to\infty} \mathbb{P}(\text{Greedy clique terminates with a clique of size at most } (1-\epsilon)\log_2(n)) = 0.$$

   [4]

Solution of Question 1.

(a) A function $f:\ \mathbb{R}^n \to \mathbb{R}$ is convex if for any given $\boldsymbol{x}_1, \boldsymbol{x}_2 \in \mathbb{R}^n$ and $\theta \in [0,1]$, it holds that

$$f\left(\theta \boldsymbol{x}_1 + (1-\theta)\,\boldsymbol{x}_2\right) \leq \theta f\left(\boldsymbol{x}_1\right) + (1-\theta)\, f\left(\boldsymbol{x}_2\right).$$

[2]

(b) For any given $\boldsymbol{x}_1, \boldsymbol{x}_2 \in \mathbb{R}^n$ and $\theta \in [0,1]$, it holds that

$$\begin{aligned} p\left(\theta \boldsymbol{x}_1 + (1-\theta)\,\boldsymbol{x}_2\right) &\leq p\left(\theta \boldsymbol{x}_1\right) + p\left((1-\theta)\,\boldsymbol{x}_2\right) \\ &= \theta p\left(\boldsymbol{x}_1\right) + (1-\theta)\, p\left(\boldsymbol{x}_2\right), \end{aligned}$$

where the first line follows from the the triangle inequality and the second line is obtained using the absolute scalability. [3]

(c)

i). The second condition for convexity is that if f is twice differentiable and its Hessian is positive semi-definite then f is convex. Note that $\frac{d^2}{(dx)^2}\left(\frac{1}{2}\left(y-x\right)^2\right) = 1 > 0$. The function $\frac{1}{2}\left(y-x\right)^2$ is convex. [3]

ii). Consider the general case first. Let f and g be convex and $h = f + g$. Then for all $\boldsymbol{x}_1, \boldsymbol{x}_2 \in \mathbb{R}^n$ and $\theta \in [0,1]$

$$\begin{aligned} h\left(\theta \boldsymbol{x}_1 + (1-\theta)\,\boldsymbol{x}_2\right) &= f\left(\theta \boldsymbol{x}_1 + (1-\theta)\,\boldsymbol{x}_2\right) + g\left(\theta \boldsymbol{x}_1 + (1-\theta)\,\boldsymbol{x}_2\right) \\ &\leq \theta f\left(\boldsymbol{x}_1\right) + (1-\theta)\, f\left(\boldsymbol{x}_2\right) \\ &\quad + \theta g\left(\boldsymbol{x}_1\right) + (1-\theta)\, g\left(\boldsymbol{x}_2\right) \\ &= \theta h\left(\boldsymbol{x}_1\right) + (1-\theta)\, h\left(\boldsymbol{x}_2\right), \end{aligned}$$

where the convexity of f and g has been used. This proves the convexity of the summation of convex functions.

More specifically, we have shown the convexity of $\frac{1}{2}\left(y-x\right)^2$. From Problem 1b, $|x|$ is convex and so is $\lambda\,|x|$. $f\left(x\right)$ is the summation of two convex functions and therefore convex. [3]

iii). It is straightforward to verify that

$$\partial f = \begin{cases} x - y + \lambda & \text{if } x > 0, \\ x - y + \lambda\,[-1, 1] & \text{if } x = 0, \\ x - y - \lambda & \text{if } x < 0. \end{cases}$$

[3]

iv). The minimiser x^* of f satisfies that $0 \in \partial f\,(x^*)$. As a result,

$$x^* = \begin{cases} y - \lambda & \text{if } y > \lambda, \\ 0 & \text{if } |y| \leq \lambda, \\ y + \lambda & \text{if } y < -\lambda. \end{cases}$$

[3]

(d) Suppose that $\boldsymbol{x}^*$ is not globally optimal, i.e., there exists a $\boldsymbol{y} \neq \boldsymbol{x}^*$ such that $f\,(\boldsymbol{y}) < f\,(\boldsymbol{x}^*)$. Consider a point $\boldsymbol{z}$ on the line segment between $\boldsymbol{x}^*$ and $\boldsymbol{y}$, i.e.,

$$\boldsymbol{z} = (1 - \theta)\,\boldsymbol{x}^* + \theta \boldsymbol{y},\ \theta \in (0, 1)\,.$$

Then it is clear that

$$f\,(\boldsymbol{z}) \leq (1 - \theta)\,f\,(\boldsymbol{x}^*) + \theta f\,(\boldsymbol{y}) < f\,(\boldsymbol{x}^*)\,,$$

where the first inequality follows from the convexity of f. As the strict inequality holds for all $\boldsymbol{z}$ on the line segment, it contradicts the fact that $\boldsymbol{x}$ is locally optimal. This proves the global optimality of $\boldsymbol{x}$. [3]

Solutions of Question 2.

(a) The matrix $\boldsymbol{A} \in \mathbb{R}^{m \times N}$ satisfies the RIP with parameters (K, δ) for $K \leq m$ and $\delta \in (0, 1)$, if for all index sets $\mathcal{I} \subset \{1, \cdots, N\}$ such that $|\mathcal{I}| \leq K$ and for all $\boldsymbol{q} \in \mathbb{R}^{|\mathcal{I}|}$, it holds that

$$(1 - \delta) \|\boldsymbol{q}\|_2^2 \leq \|\boldsymbol{A}_{\mathcal{I}} \boldsymbol{q}\|_2^2 \leq (1 + \delta) \|\boldsymbol{q}\|_2^2 .$$

The RIC δ_K is the infimum of all parameters δ for which the RIP holds, i.e.,

$$\delta_K = \inf \big\{ \delta : \ (1 - \delta) \|\boldsymbol{q}\|_2^2 \leq \|\boldsymbol{A}_{\mathcal{I}} \boldsymbol{q}\|_2^2 \leq (1 + \delta) \|\boldsymbol{q}\|_2^2 . \\ \forall\, |\mathcal{I}| \leq K, \ \forall \boldsymbol{q} \in \mathbb{R}^{|\mathcal{I}|} \big\} .$$

[4]

(b) For all $\mathcal{I} \subset \mathcal{J}$ such that $|\mathcal{I}| \leq K$ and $|\mathcal{J}| \leq K'$, we can extend $\boldsymbol{q} \in \mathbb{R}^{|\mathcal{I}|}$ to $\boldsymbol{q}' \in \mathbb{R}^{|\mathcal{J}|}$ by adding $|\mathcal{J}| - |\mathcal{I}|$ zeros to $\boldsymbol{q}$. It is clear that $\|\boldsymbol{A}_{\mathcal{J}} \boldsymbol{q}'\|_2^2 = \|\boldsymbol{A}_{\mathcal{I}} \boldsymbol{q}\|_2^2$ and $\|\boldsymbol{q}'\|_2 = \|\boldsymbol{q}\|$.

By the definition of δ_K, there exist $\boldsymbol{A}_{\mathcal{J}}$ and $\boldsymbol{q}'$ that achieve one of the bounds $(1 - \delta_K) \|\boldsymbol{q}'\|_2^2 \leq \|\boldsymbol{A}_{\mathcal{J}} \boldsymbol{q}'\|_2^2 \leq (1 + \delta_K) \|\boldsymbol{q}'\|_2^2$ arbitrarily close.

Since $\boldsymbol{q}'$ is only a *strict* subset of $\mathbb{R}^{|\mathcal{J}|}$, there may exists a $\boldsymbol{q}'' \in \mathbb{R}^{|\mathcal{J}|}$ such that $\|\boldsymbol{A}_{\mathcal{J}} \boldsymbol{q}''\|_2^2 > (1 + \delta_K) \|\boldsymbol{q}''\|_2^2$ or $\|\boldsymbol{A}_{\mathcal{J}} \boldsymbol{q}''\|_2^2 < (1 - \delta_K) \|\boldsymbol{q}''\|_2^2$. This suggests that $\delta_K \leq \delta_{K'}$. [4]

(c)

i).

$$\begin{aligned} \left\| \boldsymbol{A}_{\mathcal{I}}^T \boldsymbol{A}_{\mathcal{J}} \boldsymbol{b} \right\|_2 &= \max_{\boldsymbol{q}:\, \|\boldsymbol{q}\|_2 = 1} \left| \left\langle \boldsymbol{q}, \boldsymbol{A}_{\mathcal{I}}^T \boldsymbol{A}_{\mathcal{J}} \boldsymbol{b} \right\rangle \right| \\ &= \max_{\boldsymbol{q}:\, \|\boldsymbol{q}\|_2 = 1} \left| \boldsymbol{q}^T \boldsymbol{A}_{\mathcal{I}}^T \boldsymbol{A}_{\mathcal{J}} \boldsymbol{b} \right| \\ &\leq \max_{\boldsymbol{q}:\, \|\boldsymbol{q}\|_2 = 1} \delta_{2K} \|\boldsymbol{q}\|_2 \|\boldsymbol{b}\|_2 \\ &= \delta_{2K} \|\boldsymbol{b}\|_2 . \end{aligned}$$

[4]

ii). From the RIP property, it holds that

$$\left\| \boldsymbol{A}_{\mathcal{I}}^T \boldsymbol{y} \right\|_2 = \left\| \boldsymbol{A}_{\mathcal{I}}^T \boldsymbol{A}_{\mathcal{I}} \boldsymbol{x} \right\|_2 \geq (1 - \delta_{2K}) \|\boldsymbol{x}\|_2 .$$

From the definition of $\hat{\mathcal{I}}$, it is clear that

$$\left\|\boldsymbol{A}_{\hat{\mathcal{I}}}^T\boldsymbol{y}\right\|_2 \geq \left\|\boldsymbol{A}_{\mathcal{I}}^T\boldsymbol{y}\right\|_2 \geq (1-\delta_{2K})\left\|\boldsymbol{x}\right\|_2 > \frac{1}{2}\left\|\boldsymbol{x}\right\|_2. \tag{2.5}$$

Suppose that $\mathcal{I}\bigcap\hat{\mathcal{I}} = \phi$. Then

$$\left\|\boldsymbol{A}_{\hat{\mathcal{I}}}^T\boldsymbol{y}\right\|_2 \leq \delta_{2K}\left\|\boldsymbol{x}\right\|_2 < \frac{1}{2}\left\|\boldsymbol{x}\right\|_2,$$

which contradicts (2.5). As a result, $\mathcal{I}\bigcap\hat{\mathcal{I}} \neq \phi$. [4]

iii). It holds that

$$\begin{aligned}\left\|\boldsymbol{A}_{\hat{\mathcal{I}}}^T\boldsymbol{y}\right\|_2 &\overset{(a)}{\leq} \left\|\boldsymbol{A}_{\hat{\mathcal{I}}\bigcap\mathcal{I}}^T\boldsymbol{y}\right\|_2 + \left\|\boldsymbol{A}_{\hat{\mathcal{I}}\setminus\mathcal{I}}^T\boldsymbol{y}\right\|_2 \\ &= \left\|\boldsymbol{A}_{\hat{\mathcal{I}}\bigcap\mathcal{I}}^T\boldsymbol{y}\right\|_2 + \left\|\boldsymbol{A}_{\hat{\mathcal{I}}\setminus\mathcal{I}}^T\boldsymbol{A}_{\mathcal{I}}\boldsymbol{x}\right\|_2 \\ &\overset{(b)}{\leq} \left\|\boldsymbol{A}_{\hat{\mathcal{I}}\bigcap\mathcal{I}}^T\boldsymbol{y}\right\|_2 + \delta_{2K}\left\|\boldsymbol{x}\right\|_2,\end{aligned}$$

where (a)follows from the triangle inequality and (b) is due to the RIP. Furthermore, let $\boldsymbol{x}_{\hat{\mathcal{I}}\bigcap\mathcal{I}}$ be the sub-vector of $\boldsymbol{x}$ corresponding to $\boldsymbol{A}_{\hat{\mathcal{I}}\bigcap\mathcal{I}}$ and $\boldsymbol{x}_{\mathcal{I}\setminus\hat{\mathcal{I}}}$ be the sub-vector of $\boldsymbol{x}$ corresponding to $\boldsymbol{A}_{\mathcal{I}\setminus\hat{\mathcal{I}}}$.

$$\begin{aligned}\left\|\boldsymbol{A}_{\hat{\mathcal{I}}\bigcap\mathcal{I}}^T\boldsymbol{y}\right\|_2 &= \left\|\boldsymbol{A}_{\hat{\mathcal{I}}\bigcap\mathcal{I}}^T\left(\boldsymbol{A}_{\hat{\mathcal{I}}\bigcap\mathcal{I}}\boldsymbol{x}_{\hat{\mathcal{I}}\bigcap\mathcal{I}} + \boldsymbol{A}_{\mathcal{I}\setminus\hat{\mathcal{I}}}\boldsymbol{x}_{\mathcal{I}\setminus\hat{\mathcal{I}}}\right)\right\|_2 \\ &\overset{(a)}{\leq} \left\|\boldsymbol{A}_{\hat{\mathcal{I}}\bigcap\mathcal{I}}^T\boldsymbol{A}_{\hat{\mathcal{I}}\bigcap\mathcal{I}}\boldsymbol{x}_{\hat{\mathcal{I}}\bigcap\mathcal{I}}\right\|_2 + \left\|\boldsymbol{A}_{\hat{\mathcal{I}}\bigcap\mathcal{I}}^T\boldsymbol{A}_{\mathcal{I}\setminus\hat{\mathcal{I}}}\boldsymbol{x}_{\mathcal{I}\setminus\hat{\mathcal{I}}}\right\|_2 \\ &\overset{(b)}{\leq} (1+\delta_{2K})\left\|\boldsymbol{x}_{\hat{\mathcal{I}}\bigcap\mathcal{I}}\right\|_2 + \delta_{2K}\left\|\boldsymbol{x}_{\mathcal{I}\setminus\hat{\mathcal{I}}}\right\|_2 \\ &\overset{(c)}{\leq} (1+\delta_{2K})\left\|\boldsymbol{x}_{\hat{\mathcal{I}}\bigcap\mathcal{I}}\right\|_2 + \delta_{2K}\left\|\boldsymbol{x}\right\|_2,\end{aligned}$$

where (a) is due to the triangle inequality, (b) follows from the RIP, and (c) holds as $\left\|\boldsymbol{x}_{\mathcal{I}\setminus\hat{\mathcal{I}}}\right\|_2 \leq \left\|\boldsymbol{x}\right\|_2$.
Now recall the proved fact (2.5)

$$\left\|\boldsymbol{A}_{\hat{\mathcal{I}}}^T\boldsymbol{y}\right\|_2 \geq (1-\delta_{2K})\left\|\boldsymbol{x}\right\|_2.$$

One has

$$(1-\delta_{2K})\left\|\boldsymbol{x}\right\|_2 \overset{(a)}{\leq} (1+\delta_{2K})\left\|\boldsymbol{x}_{\hat{\mathcal{I}}\bigcap\mathcal{I}}\right\|_2 + 2\delta_{2K}\left\|\boldsymbol{x}\right\|_2,$$

where (a) is obtained by applying (2.1) and (2.2). It is then clear that

$$\left\|\boldsymbol{x}_{\hat{\mathcal{I}}\cap\mathcal{I}}\right\|_2 \geq \frac{1-3\delta_{2K}}{1+\delta_{2K}}\left\|\boldsymbol{x}\right\|_2 .$$

[4]

Solutions of Question 3.

(a)

i).

$$\begin{aligned}\mathbb{P}(w_n \geq k) &= \mathbb{P}(\exists S \subset V \text{ s.t. } |S| = k \ \& \ S \text{ is a clique}) \\ &\leq \sum_{S \subset V,\, |S|=k} \mathbb{P}(S \text{ is a clique}) \\ &= \binom{n}{k} \left(\frac{1}{2}\right)^{\binom{k}{2}} \\ &\leq n^k 2^{-k(k-1)/2} = 2^{k \log_2 n - \frac{k(k-1)}{2}},\end{aligned}$$

where the first inequality comes from the union bound. [3]

ii). For $k = (2+\epsilon)\log_2(n)$, by previous question it holds that

$$\begin{aligned}&\mathbb{P}(w_n \geq (2+\epsilon)\log_2(n)) \\ &\leq 2^{(2+\epsilon)\log_2^2 n - \frac{1}{2}((2+\epsilon)\log_2 n - 1)^2} \\ &\leq 2^{(2+\epsilon)\log_2^2 n - \frac{1}{2}\left(\left(2+\frac{1.5}{2}\epsilon\right)\log_2 n\right)^2} \\ &= 2^{-0.5\epsilon \log_2^2 n - \frac{1.5^2}{8}\epsilon^2 \log_2^2 n},\end{aligned}$$

where the first inequality comes from $k(k-1) > (k-1)^2$ and the second inequality holds as $0.5\epsilon \log_2 n > 1$ when n is sufficiently large. Let n approach infinity. Equation (2.4) is proved. [3]

(b)

i). By construction, any node in S' is connected to all other nodes and so this is the case for any pair of nodes in S'. [1]

ii).

$$\begin{aligned}q_k &= \mathbb{P}\left(\bigcup_{S \subset V,\, |S|=K} \text{algo. picks } S\right) \\ &\leq \binom{n}{k} \mathbb{P}(\text{picks } S \text{ s.t. } |S| = k) \\ &= \binom{n}{k}\left(1 - \frac{1}{2^k}\right)^{n-k},\end{aligned}$$

where $\left(1-\frac{1}{2^k}\right)$ is the probability that the algorithm picks S but fails to pick a node in $V\backslash S$ so that the node is connected to all nodes in S, and the exponent $n-k$ comes from $|V\backslash S| = n-k$. [5]

iii).

$$\begin{aligned} q_{k_0} &\leq n^{(1-\epsilon)\log_2 n}\left(1-\frac{1}{2^{(1-\epsilon)\log_2 n}}\right)^{n-(1-\epsilon)\log_2 n} \\ &\leq \exp\left\{\frac{(1-\epsilon)\log^2 n}{\log 2} - \frac{n-(1-\epsilon)\log_2 n}{n^{1-\epsilon}}\right\} \\ &= \exp\left\{-n^{\epsilon} + \frac{(1-\epsilon)\log^2 n}{\log 2} + \frac{(1-\epsilon)\log_2 n}{n^{1-\epsilon}}\right\}. \end{aligned}$$

Note that $\frac{(1-\epsilon)\log^2 n}{\log 2} + \frac{(1-\epsilon)\log_2 n}{n^{1-\epsilon}} = o\left(n^{\epsilon}\right)$ for $\epsilon > 0$. We conclude that for large n, there exists a constant $C > 0$ s.t.

$$q_{k_0} \leq \exp\left\{-Cn^{\epsilon}\right\}.$$

[4]

iv). Since q_k is increasing,

$$\begin{aligned} &\mathbb{P}\left(\text{algorithm outputs a clique of size} \leq (1-\epsilon)\log_2 n\right) \\ &\leq \sum_{k=1}^{(1-\epsilon)\log_2 n} q_k \leq (1-\epsilon)\log_2(n)\, q_{k_0} \\ &\leq (1-\epsilon)\log_2(n)\, e^{-Cn^{\epsilon}} \overset{n\to\infty}{\to} 0, \end{aligned}$$

for some $\epsilon > 0$ and $C > 0$. [4]